TreeLoom: Learning Local Ancestry from Complete Genealogies

Anonymous Authors

Abstract

Admixture leaves chromosomes as mosaics of segments inherited from different ancestral sources. Local ancestry inference (LAI) reconstructs these mosaics using relationships between query and reference genomes. Inferred genealogies reveal these relationships, but recombination changes them along the chromosome. Learning from these changing genealogies requires combining local relationships with the continuing identity of each sampled haplotype. We introduce TreeLoom, which joins graph message passing through complete trees with recurrence along the genome. Structural features condition local representations, a separate memory follows sampled haplotypes across trees, and a final recurrent network predicts ancestry. On eight European simulations reused during development, TreeLoom reaches $97.08 \pm 0.57\%$ span accuracy for modern queries, compared with $94.45 \pm 1.15\%$ for the evaluated ARGMix release and $93.78 \pm 0.84\%$ for a derived four source AncestralPaths endpoint. The matched TreeLoom-NoMem control reaches $97.00 \pm 0.50\%$ and has higher boundary F1. These results motivate learning from complete genealogies while showing why accurate genomic assignment must be assessed alongside recovery of ancestry tract boundaries.

1 Introduction

Admixed genomes record population history in the ancestry of their chromosomal segments. Local ancestry inference (LAI) assigns these segments to ancestral sources using reference genomes of known ancestry and locates the boundaries between them. Ancestry tracts can reveal when populations mixed and how genetic material moved between populations. Ancient genomes extend this record through time, allowing comparisons among individuals who lived before, during and after major changes in ancestry.

Genealogies provide a natural basis for these assignments. At a genomic position, a local tree connects sampled haplotypes, each representing one chromosome copy, through common ancestors. A query genome may share a recent branch with a reference in one region and a more distant ancestor in another. Recombination changes these relationships along the chromosome. An ancestral recombination graph (ARG) describes this changing history through a sequence of local genealogies, linking evidence within each tree to continuity along the genome.

LAI methods based on genealogies already use several parts of this structure. AncestralPaths classifies genealogical paths in analyses of ancient Eurasian ancestry (Irving-Pease et al. 2024). ARGMix combines graph representations with genomic context (Shanks et al. 2026). ARGformer learns path representations that support ancestry assignment through reference retrieval (Bonet et al. 2026). This work raises a representation question: how can a model combine the complete local genealogy with the persistent identity of each sampled haplotype as the trees change?

The distinction between sampled genomes and inferred ancestors matters. A sampled haplotype retains its identity throughout the sequence, while an inferred internal ancestor may have no reliable counterpart in a neighboring tree. Descendant groups can remain stable or change, but neither a reused node identifier nor a shared group establishes that two inferred ancestors are the same biological individual. At the same time, a change in genealogy need not imply a change in ancestry.

TreeLoom addresses these constraints using standard neural layers. Graph message passing combines information toward and away from the root of every local tree. Features of stable and changing descendant groups condition a second traversal. A separate memory carries genomic history for each sampled haplotype into this tree encoder, and a bidirectional gated recurrent unit (GRU) combines its outputs along the chromosome. The model thus distinguishes history used to form a local representation from genomic context used to classify that representation.

We evaluate TreeLoom against available pipelines based on genealogies and use the matched TreeLoom-NoMem control to examine its additional memory. The evaluation reports genomic span accuracy together with tract boundary recovery.

2 Related Work

LAI from genealogies and haplotypes. AncestralPaths distinguishes genealogical ancestry routes that can share a broad source population (Irving-Pease et al. 2024). ARGMix learns graph representations together with genomic sequence context (Shanks et al. 2026). ARGformer uses a transformer to learn genealogical path embeddings for downstream ancestry analysis (Bonet et al. 2026). FLARE estimates local ancestry directly from haplotypes (Browning, Waples, and

Browning 2023). These methods differ in representation, supervision and prediction unit. Our completed European comparison uses a common scoring endpoint, with the material release and task differences specified in the evaluation design.

Graph message passing and genomic recurrence. An upward tree traversal summarizes descendants, while a downward traversal supplies context from outside each branch. A recurrent network can then combine local representations along the chromosome. GRUs are an established mechanism for this sequential computation (Cho et al. 2014). TreeLoom combines these standard operations to retain complete local relationships and align sampled haplotypes across trees. Its contribution concerns this organization for LAI and the evidence for additional context under matched conditions.

Context across related graphs. Graph Set Transformer interleaves propagation within graphs with context from the surrounding collection (Escrig Molina, Chen, and Probst 2026). Temporal graph state space models also carry recurrent state on changing graphs (Li et al. 2024). Local genealogies supply additional structure: genomic order, unequal physical spans and persistent sampled haplotypes. These features determine which states can be linked across trees and how distance should affect the retained history.

3 TreeLoom

3.1 Prediction Task and Model Overview

Let $T_t = (V_t, E_t)$ be the inferred local tree covering genomic interval I_t, with physical length w_t. Let q index a sampled haplotype and $v_t(q)$ its node in tree t. All sampled haplotypes appear in every tree. The model predicts a probability vector p_{tq} over four source ancestries: Anatolian farmer (Ana), Caucasus hunter-gatherer (CHG), western hunter-gatherer (WHG) and eastern hunter-gatherer (EHG).

The inputs are phased observed variants, inferred tree topology and node ages, sampling ages, and known source labels for reference haplotypes. Query source labels are masked. Simulation ancestry supplies training targets and evaluation truth, while the input features, inferred trees, structural groups and prediction intervals depend only on observed data.

Figure 1 shows the model. An encoder of observed data initializes the nodes. A graph neural network (GNN) block traverses the complete tree upward and downward. Structural residuals and sample memory then modify the inputs to a second application of the same block, with shared propagation weights. The sampled node outputs remain separate, are aligned by haplotype identity across trees, and enter a final genomic GRU and ancestry classifier.

3.2 Observed Evidence and Complete Tree Traversal

A multilayer perceptron (MLP) encodes each called allele relative to observed donor and panel allele frequencies. Mean and maximum pooling summarize these features within a tree interval for each haplotype. Sampling age and the reference label, when known, complete the sampled node features. Inferred age and indicators for sampled nodes and roots describe the tree nodes, including internal ancestors. Together these features initialize a node state b_v. Learned gates use branch geometry and agreement between inferred mutations and observed alleles to scale messages between nodes.

The upward traversal combines descendant evidence. For a parent v and child u,

$$m_{u\to v} = \frac{g_u^{\uparrow} \odot W_{\uparrow} h_u}{\max(1, d_v)}, \tag{1}$$

$$s_v = \sum_{u \in \mathrm{child}(v)} m_{u\to v}, \tag{2}$$

$$h_v = \mathrm{LN}(b_v + \tanh(A_{\uparrow}[b_v, s_v] + a_{\uparrow})) . \tag{3}$$

Here d_v is the number of children, $g_u^{\uparrow}$ is the observed feature gate, $\odot$ denotes elementwise multiplication, and brackets denote concatenation. The update combines a linear projection, a hyperbolic tangent activation, a residual connection and layer normalization (LN). Division by the number of children controls the magnitude of the aggregated message.

The downward traversal supplies the context outside a child's branch. For child u, the parent context is

$$c_{v\setminus u} = o_v + b_v + s_v - m_{u\to v}. \tag{4}$$

A gated learned transformation maps this context to the child's outside state o_u, initialized to zero at the root. Subtracting the child's upward message excludes its own contribution from the returning context. The states h_v and o_v therefore summarize evidence below and outside each node. Both traversals include every inferred ancestor.

3.3 Conditioning on Local Tree Structure

Descendant sets describe tree structure without requiring correspondence between inferred ancestors. Let $C_t(v)$ contain the sampled haplotypes descended from node v. A proper clade contains at least two sampled haplotypes and excludes the complete panel. For a unary chain with the same descendant set, we use the deepest representative.

A clade is gained when absent from the preceding tree and lost when absent from the following tree. A missing neighbor gives a zero flag. Two learned residuals transform the current node state: $\Delta_v^{\mathrm{change}}$ uses gained and lost flags, and $\Delta_v^{\mathrm{local}}$ acts on clades present in consecutive trees. Neighboring trees determine these structural flags, while the transformed features come from the current tree. The exact residual definitions appear in the supplement.

The second traversal receives $b_v + \Delta_v^{\mathrm{change}} + \Delta_v^{\mathrm{local}}$ plus the sample memory contribution defined next. Reusing the first traversal's weights and observed gates propagates this conditioned input through the complete tree.

3.4 Memory for the Same Sampled Haplotype

Sample memory carries context for each sampled haplotype across genomic trees and returns that history to the tree encoder before its second traversal. It uses a bidirectional exponential moving average, distinct from the final genomic GRU.

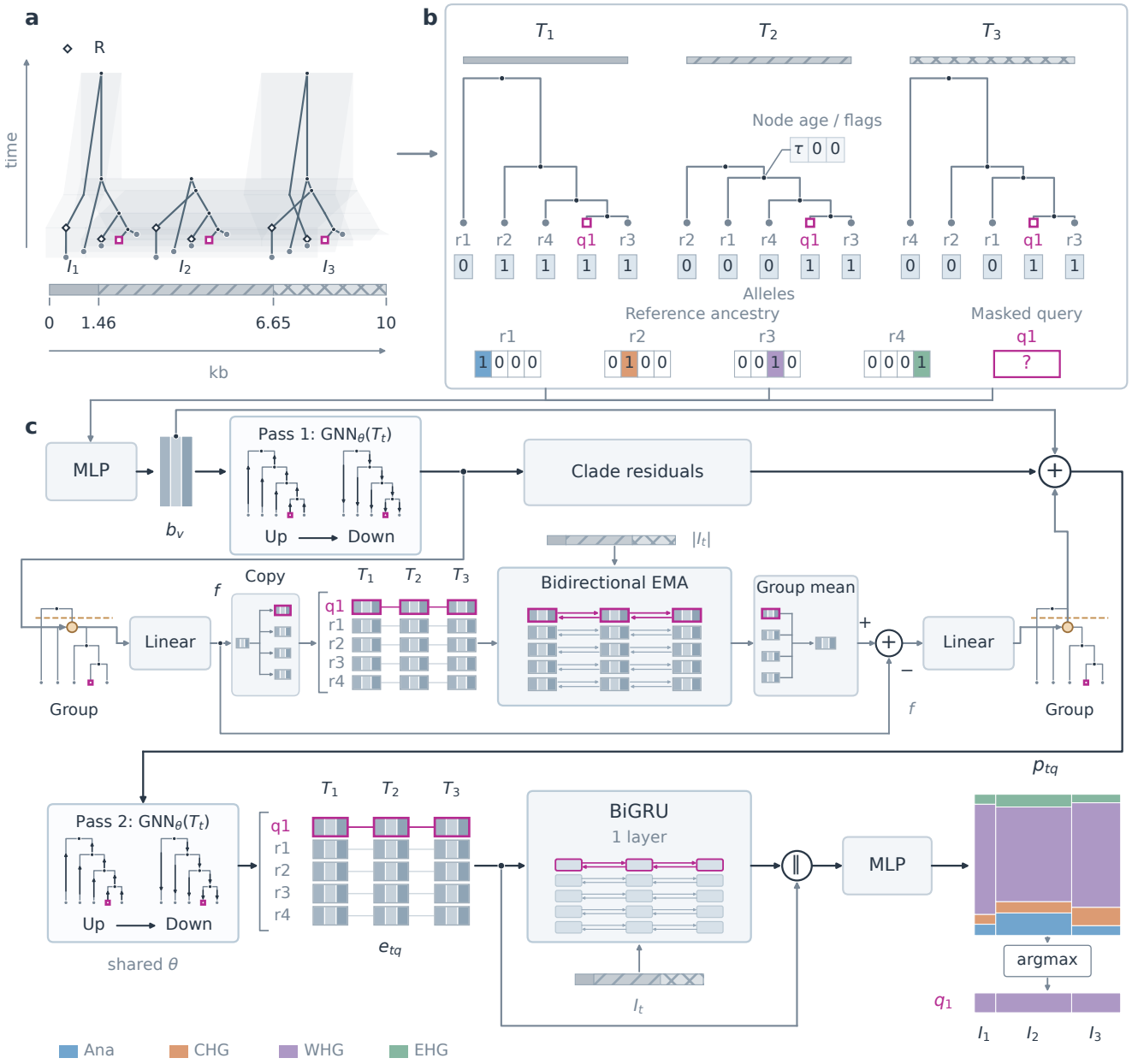


Figure 1: **a,** A recorded toy ARG and its genomic edge spans. Diamonds mark recombination. **b,** Matching marginal trees after suppressing unary nodes. Circles denote references and the magenta square query q_1. Illustrative site alleles, reference ancestry vectors, masked query ancestry (?), ages τ, and sample/root indicators form inputs. **c,** An MLP initializes node states. Each GNN pass traverses every tree upward and downward, sharing θ. Clade features produce local residuals. Highlighted nodes at a fixed age cut define groups. Their projected context f is copied to descendant sample rows. Copy and mean icons illustrate one current group's descendant rows. Bidirectional EMA follows sample identity using physical span $|I_t|$. Within each current group, the mean history minus f is projected back to the highlighted node. Corrections and original states feed pass 2. A one layer BiGRU processes aligned sample embeddings and interval geometry. Concatenation joins local and recurrent embeddings for classification. All sample rows are processed. Only query q_1 ancestry probabilities and calls are illustrated, not fitted predictions.

A fixed age cut partitions each inferred tree into groups of sampled haplotypes sharing an ancestor at the cut. Samples older than the cut retain their own sampled node. The groups depend on inferred and sampling ages. For group k in tree t, a learned projection of the representative's summed upward and outside states gives context f_{tk}. Let $k_t(q)$ denote haplotype q's group.

A forward scan carries a state for each haplotype,

$$z^{\rightarrow}_{tq} = \rho_t z^{\rightarrow}_{t-1,q} + (1-\rho_t) f_{t,k_t(q)}, \qquad \rho_t = \exp(-w_t/\lambda). \tag{5}$$

The first state equals its current context vector. A reverse scan applies the same rule in reverse genomic order. The learned positive length scale λ determines how much history is retained over a given physical span. A longer interval places more weight on its current context.

The two scan directions are averaged, then the haplotype states are averaged within each current group. Subtracting the current group context isolates the contribution of genomic history. A learned projection returns this difference as a residual at the group representative, and the second traversal propagates it through the complete tree. Each history remains attached to its sampled haplotype as group membership changes.

The matched control, TreeLoom-NoMem, removes the mechanism in Equation 5 and retains both complete traversals, structural residuals and the final genomic GRU.

3.5 Genomic Context and Ancestry Prediction

After the second traversal, a learned projection of the upward and outside states produces a local embedding $e_{tq} \in \mathbb{R}^d$ for every sampled haplotype. Aligning these embeddings by sampled identity gives one ordered sequence per haplotype.

A bidirectional GRU (BiGRU) with one layer processes the local embeddings together with interval geometry,

$$r_{1:T,q} = \mathrm{BiGRU}\left([e_{tq}, \mathrm{geometry}(I_t)]_{t=1}^{T}\right). \tag{6}$$

The geometry contains interval length and distances to neighboring interval centers. The GRU has hidden width h in each direction. A classifier combines the local embedding of width d with the resulting genomic representation of width $2h$,

$$p_{tq} = \mathrm{softmax}\left(\mathrm{MLP}([e_{tq}, r_{tq}])\right). \tag{7}$$

All sampled haplotypes and tree intervals remain present until queries are selected at readout. Training minimizes cross entropy weighted by physical span, with each simulated chromosome as one optimization example. Predictions are constant within an inferred tree interval, restricting ancestry transitions to boundaries between inferred tree intervals. Architecture widths and the training schedule are given in the supplement.

4 Evaluation Design

4.1 European Development Simulations

The European comparison uses 32 training simulations and eight validation simulations, each 2 Mb long, from the released AncestralPaths demographic formulation associated with Irving-Pease et al. (2024). The panel contains 476 diploid individuals, or 952 phased haplotypes, with 266 reference and 686 query haplotypes. Queries comprise modern individuals, Bronze Age individuals, Bronze Age Anatolians (BAA), Neolithic farmers (Neo) and Yamnaya (Yam). Simulations are split before tree intervals, rows or sequence chunks are constructed.

Relate infers genealogies from the observed variants and sampling ages (Speidel et al. 2019). All completed European pipelines use the common observed panel and inferred genealogies, represented as required by each method. The simulation assumptions and inference parameters are in the supplement. The released Bronze Age roster is only an approximate correspondence to the ARGMix preprint's Post-Neolithic category.

The eight validation simulations have been reused for model development and checkpoint selection, so the reported results are development evidence. Each condition has one optimizer seed, and independent test evaluation is unfinished. The simulation is the unit of replication because its intervals, bins and haplotypes are dependent observations.

4.2 Common Scoring Endpoint

The primary endpoint is accuracy weighted by physical span for the 182 modern query haplotypes. Secondary endpoints cover the ancient cohorts and all queries. All European predictions are scored on the same fixed 100 bp bins clipped to inferred coverage. Let m_{iqc} be the physical mass of true ancestry c for query q in bin i. Span accuracy is

$$A = \frac{\sum_{iqc} m_{iqc}\, \mathbf{1}\{c = \arg\max_k p_{iqk}\}}{\sum_{iqc} m_{iqc}}. \tag{8}$$

Each simulation receives equal weight in the reported mean. Sample SD across the eight simulations is reported in percentage points and describes simulation variation for a fitted model. It does not quantify variation across training seeds or form a confidence interval.

Accuracy measures the amount of correctly assigned sequence. Boundary precision, recall and F1 assess whether the ancestry transitions are located correctly. Boundaries are matched one to one within 1 kb. The matcher uses position and does not require the predicted transition to have the correct pair of source labels. Negative log likelihood (NLL) uses the same physical masses and the original predicted probabilities, without flooring or recalibration.

4.3 Comparator Scope and Matched Memory Control

The AncestralPaths comparison uses the released dense six output classifier. We sum its six path probabilities into the four source probabilities before class assignment,

$$(p_{\mathrm{Ana}}, p_{\mathrm{CHG}}, p_{\mathrm{WHG}}, p_{\mathrm{EHG}}) = (p_1 + p_5, p_2 + p_6, p_3, p_4). \tag{9}$$

This derived endpoint differs from the native six path task. The paper describes a convolutional classifier, while the released implementation uses a dense classifier. Its bounded replacement sampler also leaves 9,126 requested BAA path six training rows unfilled, with an unmeasured effect on accuracy. The five epoch author schedule is completed and its final epoch is used.

ARGMix uses the current author release, with 15 references and no ancestor inputs. The preprint describes 25 references and three inferred ancestors. The public release does not provide a supported implementation of that earlier input

Pipeline	Accuracy ↑	Boundary F1 ↑
AncestralPaths	93.78 ± 0.84	7.03 ± 0.78
ARGMix, current release	94.45 ± 1.15	11.74 ± 1.73
TreeLoom	$\mathbf{97.08 \pm 0.57}$	$\mathbf{18.38 \pm 1.31}$

Table 1: Modern European development performance, mean±sample SD in percent across eight 2 Mb simulations and 182 query haplotypes. AncestralPaths probabilities are collapsed from six paths to four source classes. ARGMix uses the current author release. Boundary matching uses a 1 kb positional tolerance. Bold marks the highest mean. Inputs, training and selection differ between pipelines as detailed in the text.

configuration. We adapted reference slots and metadata so the released model could process the common sample panel. The released model, training augmentation and stopping rule are retained. ARGMix is trained for 11 epochs. Its own validation criterion selects epoch nine.

TreeLoom also receives observed single nucleotide polymorphism (SNP) features. The methods differ in reference eligibility, representation, supervision unit, training diversity, cumulative optimization and checkpoint selection. The common scorer compares the fitted pipelines on this endpoint, without equating their inputs or training budgets. These conditions are not exact publication reproductions and do not establish an overall LAI ranking. Full training and release differences are retained in the supplement.

The matched comparison asks whether adding sample memory changes performance under the specified training procedure. TreeLoom and TreeLoom-NoMem start from the same trained local encoder, receive 960 additional updates with all weights trainable, and use the same checkpoint rule. The memory arm has 513 additional parameters, and each arm selects its checkpoint independently. Earlier stages with different cumulative training exposure are described only as training lineage in the supplement.

5 Results

5.1 Span Accuracy on the Shared European Endpoint

TreeLoom reaches $97.08 \pm 0.57\%$ modern span accuracy (Table 1). The evaluated current ARGMix release reaches $94.45 \pm 1.15\%$, and the released dense AncestralPaths classifier reaches $93.78 \pm 0.84\%$ on the derived four source endpoint. TreeLoom assigns 2.63 percentage points more sequence correctly than this ARGMix condition.

The fitted TreeLoom pipeline therefore assigns more genomic span correctly on this endpoint. The matched comparison below addresses the smaller difference associated with adding sample memory.

5.2 Accuracy across Modern and Ancient Cohorts

TreeLoom has the highest measured accuracy in each listed cohort (Table 2). The cohorts represent different sampling ages, mixtures and tract structures. Small groups such as BAA and Yam provide limited evidence about individual ancestry transitions.

5.3 Added Memory Changes Accuracy and Boundaries Differently

TreeLoom-NoMem reaches $97.00 \pm 0.50\%$ accuracy, only 0.08 percentage points below TreeLoom (Table 3). Its boundary F1 is higher, at $19.05 \pm 1.43\%$ compared with TreeLoom's $18.38 \pm 1.31\%$. The paired mean changes after adding sample memory are therefore $+0.08$ accuracy points and -0.67 boundary F1 points.

The two metrics answer different biological questions. Correctly assigned genomic mass describes broad ancestry patterns, while missed, extra or displaced transitions alter the inferred tracts. The memory arm has slightly higher span accuracy and lower boundary F1 under this training procedure. TreeLoom denotes the condition selected by modern span accuracy.

Additional matched controls in the supplement show limited benefit from more elaborate exchange between trees.

5.4 High Span Accuracy Leaves Substantial Tract Error

Boundary F1 remains low for all three completed pipelines. TreeLoom reaches 18.38%, despite assigning 97.08% of the represented genomic span correctly. This gap shows that a small amount of incorrectly assigned sequence can coexist with many missed or extra ancestry transitions. Span accuracy alone is therefore insufficient evidence that ancestry tracts are recovered faithfully.

The readout can change ancestry only at boundaries between inferred tree intervals, which need not resolve the true transitions. A readout with more flexible transition locations and a training objective that rewards boundary fidelity are therefore directions for future work.

6 Discussion

TreeLoom combines descendant and outside context within each genealogy with continuity along the chromosome. Its two routes for genomic history act before and after the second tree traversal. Both follow persistent sample identities without assuming correspondence between inferred ancestors.

On the European development endpoint, TreeLoom assigns more genomic span correctly than the evaluated AncestralPaths and ARGMix pipelines. The comparison supports the fitted representation under the specified demographic formulation. Material input and training differences prevent attribution of the gap to one neural component, and reused validation simulations cannot establish generalization to unseen data. Independent simulations, multiple optimizer seeds and completion of the native comparisons remain necessary to assess relative method strength.

The matched memory result is narrower: slightly higher span accuracy accompanies lower boundary F1. Additional context from repeated clades and alternative memory transport rules also yield limited benefit in their matched controls. These findings focus attention on tract recovery. Even a

Cohort	Query haplotypes	AncestralPaths	ARGMix	TreeLoom
Modern	182	93.78 ± 0.84	94.45 ± 1.15	$\mathbf{97.08 \pm 0.57}$
Bronze Age	162	94.76 ± 1.18	97.71 ± 0.75	$\mathbf{99.16 \pm 0.43}$
BAA	14	96.94 ± 1.32	98.44 ± 1.02	$\mathbf{99.35 \pm 1.04}$
Neo	302	97.57 ± 0.27	99.30 ± 0.26	$\mathbf{99.73 \pm 0.07}$
Yam	26	95.43 ± 1.41	99.22 ± 0.90	$\mathbf{99.94 \pm 0.12}$
Ancient	504	96.54 ± 0.32	98.76 ± 0.31	$\mathbf{99.55 \pm 0.13}$
All queries	686	95.81 ± 0.45	97.61 ± 0.50	$\mathbf{98.89 \pm 0.23}$

Table 2: Cohort span accuracy in percent, mean±sample SD across the same eight development simulations. AncestralPaths uses the derived four source endpoint, and ARGMix uses the current author release. Bold marks the highest mean per row. Ancient and all queries aggregate the corresponding cohorts above them. The Bronze Age roster is not established as identical to the ARGMix paper's Post-Neolithic panel. Cohorts differ in sampling age, ancestry mixture and tract structure, so high simulated ancient accuracy does not establish robustness to empirical ancient DNA.

Matched model	Accuracy ↑	Boundary F1 ↑
TreeLoom-NoMem	97.00 ± 0.50	$\mathbf{19.05 \pm 1.43}$
TreeLoom	$\mathbf{97.08 \pm 0.57}$	18.38 ± 1.31

Table 3: Effect of added sample memory after the same initialization and additional training budget. Values are mean±simulation SD in percent on the same modern endpoint. TreeLoom-NoMem retains the final genomic GRU and both complete tree traversals. Checkpoints are selected separately within each arm. Bold identifies the higher mean per metric.

model that assigns most bases correctly can miss or add many transitions, so ancestry evaluation needs both measures.

The ancient cohorts further illustrate the role of biological assumptions. Their sampling histories are explicit in simulation, while empirical ancient genomes introduce uncertainty in phasing, genotype calls, reference representation and genealogical reconstruction. The supplement retains a descriptive application of an earlier TreeLoom condition together with genotype affinity counterevidence. Independent evidence of local ancestry is needed to validate such empirical predictions.

Code availability. Source code, experiment configurations and complete results are available at https://github.com/alexeyshmelev/ARG-GNN.

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Supplementary Material

A Simulation Assumptions and Parameters

The European simulations adopt the released AncestralPaths demographic formulation associated with Irving-Pease et al. (2024). The numerical assumptions below define this study condition. They are not demographic estimates inferred from the simulated observations. A census at 350 generations defines four source ancestry supervision before the older mergers of source populations. This truth census is distinct from the model's observed tree cut at the same nominal time.

The panel has 343 query and 133 diploid individuals from pure reference populations. Queries comprise 91 modern, 81 Bronze Age, seven BAA, 151 Neo and 13 Yam diploids. Pure references comprise 51 WHG, 43 EHG, 25 Ana and 14 CHG diploids. Modern samples have age zero, and ancient samples retain the released age roster in generations. The resulting 952 input haplotypes include 686 queries.

Time (generations)	Descendant	Ancestral proportions
166	Present/Bronze	0.5 Neo, 0.5 Yam
169	BAA	0.25 CHG, 0.75 Ana
177	Yam	0.5 CHG, 0.5 EHG
259	Neo	0.25 WHG, 0.75 Ana
600	WHG, EHG	Shared NE ancestor
800	CHG, Ana	Shared WA ancestor
1,500	NE, WA	Common trunk

Table 4: European demographic events adopted from the released formulation associated with AncestralPaths (Irving-Pease et al. 2024). Times are measured backward from the present. Proportions specify ancestral contributions at the corresponding event.

Effective population sizes are 50,000 for Present/Bronze and Neo, 20,000 for Yam and BAA, 3,000 for EHG and CHG, 2,000 for WHG, and 5,000 for Ana. The older NE, WA and trunk populations each have size 15,000. Growth and migration between events are zero. The simulation uses the Hudson coalescent, discrete genomic positions, a uniform recombination rate of 10^{-8} per base per generation, and JC69 mutations at 1.25×10^{-8} per base per generation (Baumdicker et al. 2022). The 2 Mb sequence length and 32/8 training/validation split define the present development panel.

Relate uses the observed biallelic variants and sampling ages (Speidel et al. 2019). Its initial effective haploid size is 30,000, mutation rate is 1.25×10^{-8} and generation time is 28 years. The European inference includes five iterations of population size estimation followed by estimation of branch lengths. All applicable completed model adapters consume this same final reconstruction. Mutation agreement features use observed alleles on the inferred trees.

The demographic events and population sizes follow the released AncestralPaths simulator and the Supplementary Information of Irving-Pease et al. (2024), particularly Figure S7 and pages 25, 27–28 and 30–32. Pulse times in the release are one generation older than the oldest sample in each corresponding admixed cohort. This gives 169 generations for BAA, whereas the paper reports 170. We use the released value. The basal merger is fixed at 1,500 generations as in the code. The source supplement also gives 15,000 years for that event, an unresolved difference in units. These are adopted demographic assumptions.

The complete released roster supplies the sampling ages of all 476 diploid individuals. Modern ages are fixed at zero. Ancient ages retain their individual values rather than cohort means. Table 5 summarizes their ranges. The released example uses 2 Mb sequences, while the AncestralPaths paper describes 200 Mb sequences, three training simulations and five evaluation simulations. Our 32/8 split is a separate development design.

Cohort	Diploid individuals	Age range, generations
Modern	91	0
Bronze Age	81	100–165
BAA	7	136–168
Neo	151	139–258
Yam	13	153–176
WHG	51	210–529
EHG	43	155–402
Ana	25	217–360
CHG	14	184–473

Table 5: Complete simulation panel summarized from the released AncestralPaths age roster. The first five cohorts are queries. The last four are pure source references. Ages are retained for each individual, and both phased haplotypes share that individual's sampling age.

Our simulations introduce no additional genotype error. The map is synthetic and uniform at 1 cM/Mb. The sequence occupies $[0, 2{,}000{,}000)$ in coordinates starting at zero. Variant positions are shifted by one for VCF input and shifted back after inference. Scoring uses fixed 100 bp bins and clips them to inferred coverage, leaving uncovered terminal sequence unscored. The chromosome label used for interchange does not identify an empirical chromosome or genome assembly. Complete individual ages and the fixed ancestry, mutation and inference seeds are included with the released experiment configurations.

The April ARGMix benchmark instead uses 1,611 individuals, a chromosome 22 recombination map, genotype error at rate 0.001, admixed informative references and demographic perturbations (Shanks et al. 2026). It excludes BAA. Our unperturbed 476-person condition therefore supports a comparison of fitted pipelines on common data, rather than replication of that benchmark.

Scientific TODO. Resolve the April ARGMix empirical map identity and coordinate domain, complete individual age roster, genotype error mechanism, reference eligibility and demographic perturbation draws before reproducing its simulation experiment. For the American benchmark, provide a primary source demographic table and resolve the haplotype interpretation and focal coordinate assumptions.

B Model and Training Parameters

B.1 Structural Residual Definitions

The residuals use the descendant sets and representative nodes defined in Section 3.3.

A node's gained flag g_v is one when its proper clade is absent from the preceding tree. Its lost flag ℓ_v is one when the clade is absent from the following tree. Flags at the missing sequence neighbors are zero. The flags condition a local residual,

$$\Delta_v^{\mathrm{change}} = O[g_v(Uh_v \odot Vh_v) - \ell_v(Uh_v \odot Wh_v)]\,. \quad (10)$$

The linear projections pass through a lower dimensional space. A second residual acts on clades present in at least two consecutive trees,

$$\Delta_v^{\mathrm{local}} = \chi_v O_l[(U_l h_v) \odot (V_l h_v) + W_l h_v]\,, \quad (11)$$

where χ_v marks eligible clades. Both residuals transform the current node representation. Neighboring trees determine the structural flags, while the features being transformed come from the current tree.

B.2 Architecture and Optimization

The genetic site encoder has two hidden layers of width 32. For a called allele it receives donor log frequency features, corresponding availability indicators, panel frequency, site entropy and a call indicator. Eight donor groups combine the four reference ancestries with ages below or at least 350 generations. Allele frequencies use additive pseudocount 0.5. If a represented reference class has no called alleles at a site in an age group, all represented classes in that age group are masked for that site. Mean and maximum pooling are combined with call count, missing pool status and interval span.

The inferred tree cut for sample memory is fixed at 350 generations. Samples older than the cut retain their own sampled node. Tree states have width 32 and local residual projections have rank eight. The memory projection also has rank eight and adds 513 parameters. Its length scale starts at 5 kb and is constrained to 100 bp–100 kb. Residual output projections start at zero. The genomic GRU has one layer and width 32 in each direction. The ancestry classifier receives 96 features, maps through width 32 and produces four scores. TreeLoom has 32,325 parameters. TreeLoom-NoMem has 31,812.

Training minimizes physical span cross entropy over all 686 queries,

$$\mathcal{L} = -\frac{\sum_{tqc} M_{tqc} \log p_{tqc}}{\sum_{tqc} M_{tqc}}, \qquad M_{tqc} = \sum_{i:r(i)=t} m_{iqc}. \quad (12)$$

Here $r(i)$ maps a scored bin to its inferred tree interval. Summing target masses within an interval preserves the original objective because the model's prediction is constant there. A complete simulated chromosome is one optimization example.

AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, betas $(0.9, 0.999)$ and epsilon 10^{-8}. Gradient norm is clipped at one. The reported seed is 101. The shared initializer is trained over three successive stages of 30 passes on the 32 training simulations, totaling 2,880 updates as tree processing and local conditioning are introduced. These successive stages are a training lineage, not matched component ablations. Optimizer moments reset at each stage.

For the final comparison, both TreeLoom and TreeLoom-NoMem start from the same model trained for 2,880 updates and complete 30 further passes, or 960 new updates. All inherited weights remain trainable. Checkpoints every five passes are ranked by equal simulation modern validation accuracy, then lower NLL, then earlier pass. TreeLoom selects added pass 30, or 3,840 cumulative updates. TreeLoom-NoMem selects added pass 20, or 3,520 cumulative updates, while its run also completes 3,840 updates. The alternate memory rules below share the same initializer, capacity and completed budget as TreeLoom.

C Comparator Conditions and Unresolved Recipe Differences

AncestralPaths. The executed archive contains a flattening operation on a 5×9 representation, a dense layer of width 512 with ReLU, dropout 0.2 and six softmax outputs. This differs from the convolutional classifier described in the paper (Irving-Pease et al. 2024). The fit completes five epochs with Adam learning rate 0.001, batch size 32 and shuffling. A deterministic seed of 101 is supplied by the adaptation. The final epoch is used without validation checkpoint selection. Modern deterministic compatibility does not establish numerical equivalence to the original legacy execution.

The bounded sampler accepts 5,190,874 of 5,200,000 requested rows with replacement. All 9,126 missing rows are BAA path six examples, amounting to 2.2815% of requested BAA rows and 0.1755% of all requested rows. There are 53,214 replacement duplicates. Other cohort quotas are met. The sampler stops after no progress or a bound of 128 passes, so its realized shortfall is a material adaptation whose accuracy effect has not been measured. Genomic training diversity is 32×2 Mb here, whereas the paper describes 3×200 Mb. Completing five epochs does not establish convergence, and training loss is still decreasing at epoch five.

The four source probability collapse in the main text is separate from six path prediction, a retrained four path task, or evaluation restricted to non BAA routes. It must not be compared directly with a published native six path accuracy. Strict NLL is infinite in the present derived endpoint because a true source receives zero probability over 800 query base pairs. No probability floor is applied.

ARGMix. The evaluated author release uses 15 references without inferred ancestor nodes. Its transformer, bidirectional GRU and native training procedure are retained (Shanks et al. 2026). It uses chunks of 1,024 rows with stride 512, yielding 21,952 training and 5,488 validation chunks. All 686 queries are available, with no local subgraph cap. Training drops the nearest reference, validation drops none, label smoothing is 0.1 and loss weighting is absent. Source seed 50 is retained. The model uses supervision at the first SNP of an interval and averages overlapping softmax predictions arithmetically.

Its native selection objective is the smoothed, masked and unweighted mean validation batch loss. The selected epoch is nine of 11 completed epochs, with patience two and a cap of 20 epochs. The external span scorer does not replace this inner selection rule. The model with 25 references and three ancestor inputs described in the preprint is not recovered by the current release, whose supported loaders and extraction reject ancestor inputs. A current release run with 25 references would still have zero ancestors and would be a separate sensitivity condition. The original model, exact reference eligibility, checkpoint and publication evaluator remain unresolved. Differences in the observed panel, map, error process, training diversity and prediction support also remain material.

American comparators. A separate American benchmark uses a 30 Mb simulation with 600 reference and 1,000 query haplotypes from EUR, EAS and AFR sources. Native FLARE calls score 98.0100% accuracy and 97.7151% macro F1 on 10,000 predetermined focal records from one validation simulation. This metric counts focal records rather than physical span and has no between simulation SD. It therefore cannot be compared numerically with the European span endpoint or establish a comparative ranking on its own.

The completed FLARE condition uses author release v0.6.0 with native defaults: minimum minor allele count 50, minimum minor allele frequency 0.005, expectation maximization enabled, ten generations, ancestry proportion updates disabled, array mode disabled and seed -99999. Native filtering retains 27,558 of 228,477 observed markers. The evaluator uses original hard ancestry calls at the nearest native marker, with midpoint boundaries, on all 10,000 fixed focal records. It does not score a filtered subset chosen after the outcome. All prediction identities and coordinates are checked before private truth is opened.

ARGformer's completed American ancestry pipeline remains unavailable. Its published PCA/clustering classifier and nearest reference retrieval are distinct downstream routes (Bonet et al. 2026). Exact clustering, class mapping and American retrieval settings are not recovered. The prepared route retains the author release's learned normalized embedding and uses nearest reference retrieval as a disclosed reconstruction. The paper states contrastive temperature 0.05 and inverse frequency anchor weighting, whereas the release configuration uses 0.30 and its stock route can use uniform weights. These differences remain unresolved. Pretraining alone does not supply ancestry accuracy.

Scientific TODO. Complete independent European tests, additional optimizer seeds and larger panels. For publication reproduction, recover the original ARGMix model, reference roster, map, genotype error settings and evaluator, plus the stated AncestralPaths convolutional model and evaluator. Assess the AncestralPaths sampling deficit and ARGMix reference restriction. Complete the matched American comparison using common observations, reference and query identities, focal records and truth support, resolving its demographic and ploidy reconstruction and the published ARGformer downstream evaluator and objective differences.

Conditioning context	Accuracy ↑	Boundary F1 ↑
Own node	96.82 ± 0.61	18.06 ± 1.78
Other episode members	$\mathbf{96.83 \pm 0.59}$	$\mathbf{18.16 \pm 1.38}$

Table 6: Matched test of context from other trees at an earlier shared initialization, with 960 added updates per arm. Values are mean±simulation SD in percent on modern queries. Both arms select pass 30. This is a conceptual context comparison, not a comparison of final training budgets with TreeLoom.

Memory rule	Accuracy ↑	Boundary F1 ↑
Group transport	97.05 ± 0.58	18.85 ± 1.05
Intersection transport	97.06 ± 0.56	18.93 ± 1.42
TreeLoom (sample memory)	$\mathbf{97.08 \pm 0.57}$	18.38 ± 1.31
TreeLoom-NoMem	97.00 ± 0.50	$\mathbf{19.05 \pm 1.43}$

Table 7: Matched memory rules, with the same initialization and 960 added updates per arm. Values are mean±simulation SD in percent. The first three arms each have 32,325 parameters and select added pass 30. The control has 31,812 parameters and selects pass 20. Bold identifies the largest mean per metric.

D Additional Scientific Results

D.1 Limited Matched Tests of Additional Context

A matched test replaces the own node context in Equation 11 with a span weighted average from the other trees in the same consecutive proper clade episode. For member i of episode $E(i)$,

$$\bar{h}_{-i} = \frac{\sum_{j \in E(i)} w_j h_j - w_i h_i}{\sum_{j \in E(i)} w_j - w_i}. \tag{13}$$

Only episodes with at least two trees are eligible. Replacing $V_l h_i$ with $V_l \bar{h}_{-i}$ leaves the parameter count, eligible nodes, initialization and added training budget unchanged. This comparison tests the value of information from other trees beyond an extra local transformation.

The paired accuracy difference is 0.0096 ± 0.1527 percentage points. The paired boundary F1 difference is 0.1001 ± 0.9632 points. The accuracy gain falls below the registered practical threshold of 0.10 points.

We also compare sample memory with two alternatives of the same capacity. Group transport pools history through the intermediate cut group when group membership changes. Intersection transport retains the intersections of consecutive groups as the units of transport. Both use the same context from the first traversal, distance weighting, second traversal and final genomic readout as sample memory.

Intersection transport changes accuracy by $+0.0137 \pm 0.0243$ points relative to group transport and by -0.0138 ± 0.0384 points relative to sample memory. It fails both registered practical criteria against every control: at least 0.10 accuracy points gained without mean boundary F1 loss, or at least one boundary F1 point gained with at most 0.05 accuracy points lost. These are development utility thresholds, not statistical significance tests.

D.2 Descriptive Application to Ancient Danish Genomes

An earlier TreeLoom condition with context from other members of a clade episode was applied without further fitting to ten ancient Danish query individuals and eight proxy reference individuals from the imputed genomes analyzed by Irving-Pease et al. (2024). This is the episode condition in Table 6, not the final model with sample memory. Five earlier individuals date to 5.44–5.67 ka cal BP and five later individuals to 3.97–4.26 ka cal BP. Three fixed 10 Mb windows on chromosomes 20–22 contain 1,158 inferred trees and 8,587 qualified site records.

Assignments weighted equally across people integrated over physical span are 14.27 percentage points higher for EHG and 5.52 points higher for CHG in the later group. EHG has a positive contrast on each chromosome. CHG depends on chromosome 21 and changes to -0.68 points when it is omitted. Separate genotype affinity contrasts from the same imputation resource do not support the registered directions. There is no local ancestry truth or calibration of the model's outputs as mixture fractions. These descriptive predictions therefore illustrate application to empirical genealogies and do not validate the historical interpretation.

D.3 County Forecasting with the Adapted Core

We adapt TreeLoom's two shared GNN passes, memory and recurrent core to forecast UCI Hungarian Chickenpox Cases[1]. Six 40 epoch fits, with three initialization seeds per arm, total 4,800 updates. Forecasts use 52 observed weeks and validation MAE selection. TEST covers 20 counties and 79 weeks.

Condition	MAE ↓	RMSE ↓
TreeLoom core	13.908 ± 0.243	$\mathbf{24.269 \pm 0.094}$
NoMem core	$\mathbf{13.648 \pm 0.051}$	24.408 ± 0.106
Last value	17.497	31.412
Seasonal lag 52	20.022	33.840
Ridge ($\alpha = 1000$)	18.959	27.298

Table 8: Errors in cases, mean±sample SD across seeds. Bold marks the lowest mean.

Memory fails the prespecified 2% MAE and paired uncertainty criteria despite lower RMSE (Table 8). Paired Full minus NoMem MAE is $+0.260$ cases, with a conditional 95% interval $[0.105, 0.427]$ from 10,000 circular resamples of 13 weeks on this fixed graph and series. Shuffling the first 51 history weeks while fixing the latest week raises both MAEs, supporting reliance on order.

[1] https://doi.org/10.24432/C5103B.